

Electrical and Computer Science engineers, even Chemical engineers or someone with a strong background in Physics. As for degree certification, companies hire all the way up from a Bachelors degree to a Masters in Science degree, as well as PhD holders; thus, spanning the entire spectrum of educational degrees.

Technical requirements

A strong hardware engineering background with specialisation in VLSI design is a must. Additionally, knowledge of digital systems and VLSI design, computer architecture, computer arithmetic, C/C++ programming languages, CMOS transistors and circuits is necessary. In VLSI design, tools play a major role. An integral part of an engineer's work deals with using the latest industry tools from Synopsys, Cadence and other EDA vendors. An understanding of hardware languages such as Verilog is also important. Other languages such as Perl and C come in handy too. Processor description languages such as nML and Lisa will be an added plus if you're thinking of a career specialisation in microprocessor design.

"Apart from architecture and HDLs, chip designing in today's world will need in-depth knowledge of even transaction level modeling using SystemC, knowledge of electronic design automation tools, and some understanding of the backend timing / manufacturing issues. Knowledge of simulation tools and interactions with memory/cache design will also be needed." says Sandeep Kumar, a consultant architect at Coware India Pvt Ltd. Sandeep has sketched a successful career in this field, now spanning over 17 years of experience.

Recruitment

Companies are in search of highly motivated individuals with strong analytical and communication skills. Apart from educational degrees, companies are also interested in the fundamentals. "There are people who understand how to push the tools, but when we talk to candidates we try to see if they are familiar with fundamentals. For example, if we're working on a GCC component, we would want to know where you put this particular switch. Why here? Why do you use it? OK, you know how to use ASIC tools, but do you know why you did certain things that way? Ability to understand and pick up is important," said Karthik. Arriving at the answer is as important as getting to the answer itself. So you can expect these kinds of questions in the interview rounds. Most companies have tests for analytical ability.